

Example 2: Find the standard matrix A for the dilation transformation $T(x) = 3x$, for x in $\mathbb{R}^2$.

Solution: $T(e_1) = 3e_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$ and $T(e_2) = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$

By Theorem 10, $A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$. $\square$

Example 3: Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the transformation that rotates each point in $\mathbb{R}^2$ about the origin through an angle φ , with counterclockwise rotation for a positive angle. We could show geometrically that such a transformation is linear. Find the standard matrix A of this transformation.

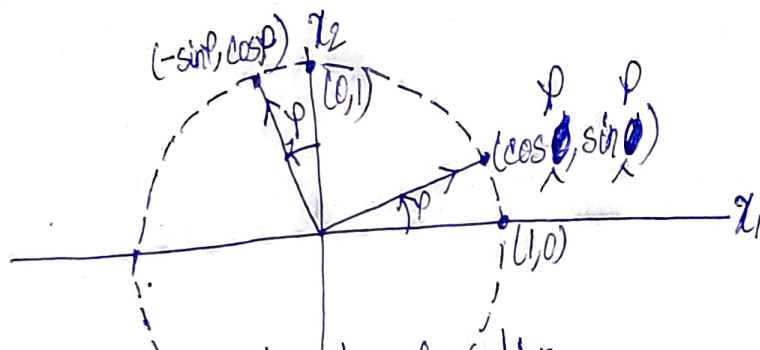


Figure 1: A rotation transformation

Solution: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ rotates into $\begin{bmatrix} \cos \varphi \\ \sin \varphi \end{bmatrix}$, and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ rotates into $\begin{bmatrix} -\sin \varphi \\ \cos \varphi \end{bmatrix}$.

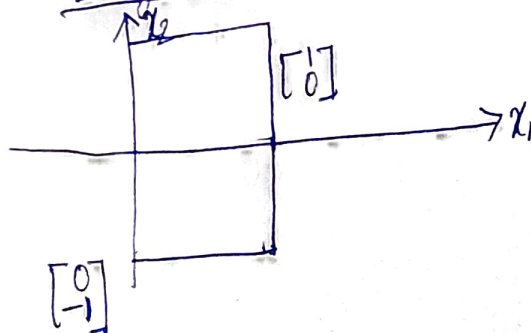
By Theorem 10, $A = \begin{bmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{bmatrix}$. $\square$

Geometric Linear Transformations of $\mathbb{R}^2$

Table 1 Reflections

Transformations
Reflection through
the x_1 -axis

Image of the Unit Square



Standard Matrix

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$